NPTEL Online Certification Courses

Indian Institute of Technology Kharagpur

Urban Informatics

Assignment - Week 7

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **12** **Total marks: (4 X 1) + (8 X 2) = 20 marks**

MCQ/MSQ Question

QUESTION 1: [1 mark]

Which of the following Python library is used in building, training, and evaluating machine-learning models?

a. Matplotlib
b. NumPy
c. Sci-kit learn
d. Pandas

Correct Answer: c. Sci-kit learn

Detailed explanation: Scikit-learn is a Python library that provides a simple and efficient framework for building, training, and evaluating machine-learning models, supporting both supervised and unsupervised learning tasks. *(Refer to lecture 31, slide 7.)*

QUESTION 2: [2 marks]

Consider the following statements regarding **Supervised Learning**.

Statement A - Supervised learning uses unlabeled data to learn predictive relationships.
Statement B - In supervised classification learning, the set of classes are mutually exclusive and collectively exhaustive.
Statement C - Supervised regression is a supervised learning task in which, given an input feature vector, the goal is to assign the example to one of N predefined classes.
Statement D – The prediction phase of supervised learning uses training dataset to test the accuracy of the model.

Based on the statements, which of the following option is **CORRECT**?

a. Both Statement A and Statement D are TRUE.
b. Only Statement B is TRUE.
c. Statement A is TRUE but Statement B is FALSE.
d. Only Statement C is TRUE.

Correct Answer: b. Only Statement B is TRUE.

Detailed explanation: Supervised learning uses historical labeled data to learn predictive relationships. In supervised classification learning, the set of classes are mutually exclusive and collectively exhaustive. Supervised classification is a supervised learning task in which, given an input feature vector, the goal is to assign the example to one of N predefined classes.

The prediction phase of supervised learning uses testing dataset to test the accuracy of the model. *(Refer to lecture 31, slide 4,5 & 6.)*

<u>QUESTION 3:</u> [2 marks]

Match the following **Modules** with their **Uses**.

Module	Use
P. sklearn.model_selection	i. used for filling or estimating missing values.
Q. sklearn.impute	ii. used to prepare raw data for ML algorithms.
R. sklearn.feature_selection	iii. used for splitting datasets into training, validation and test sets.
S. sklearn.preprocessing	iv. used to reduce dimensionality by selecting relevant features

a. P-i; Q-iii; R-ii; S-iv
b. P-ii; Q-i; R-iv; S-iii
c. P-iii; Q-iv; R-ii; S-i
d. P-iii; Q-i; R-iv; S-ii

Correct Answer: d. P-iii; Q-i; R-iv; S-ii

Detailed explanation: *Refer to lecture 31, slide 7*.

For Questions 4,5, & 6 refer to the case scenario given below.

The transport authority of a medium-sized city wants to build a model for predicting if bus stop is needed for a certain area or not. In order to build the model, they collected the following dataset:

Area ID	*Population Density*	*Traffic Volume*	*Commercial Activity*	*Existing Public Transport*	*Bus Stop Needed (Target)*
1	High	High	High	No	Yes
2	High	High	Low	No	Yes
3	High	Medium	High	Yes	Yes
4	High	Low	High	No	Yes
5	Medium	High	High	No	Yes
6	Medium	Medium	High	Yes	Yes
7	Medium	Medium	Low	No	Yes
8	Medium	Low	Low	Yes	No
9	Medium	Low	High	Yes	No
10	Low	High	Low	No	Yes
11	Low	Medium	Low	Yes	No
12	Low	Low	Low	Yes	No
13	Low	Low	High	Yes	No
14	High	High	High	Yes	Yes
15	High	Medium	Low	No	Yes
16	Medium	High	Low	Yes	Yes
17	Medium	Low	Low	No	No
18	Low	Medium	High	No	Yes
19	Low	High	High	Yes	Yes
20	High	Low	Low	Yes	No

You are appointed to assist the authorities in training a decision tree model with the given dataset. Answer the following questions based on the information given above.

<u>QUESTION 4:</u> [2 marks]

What is the total entropy of the entire dataset?

a. 0.971
b. 0.852
c. 0.934
d. 0.916

Correct Answer: c. 0.934

Detailed explanation: *Refer to lecture 33, slide 11*.

$$Entropy = -\sum_{i=1}^{n} p_i log_2(p_i)$$

Total number of observations = 20
Total Number of observations with Yes for target variable (Bus stop needed) = 13
Total Number of observations with No for target variable (Bus stop needed) = 7

$$Entropy_{total} = -\left[\left\{\frac{13}{20} log_2 \frac{13}{20}\right\} + \left\{\frac{7}{20} log_2 \frac{7}{20}\right\}\right] = -[\{0.65 \times log_2 0.65\} + \{0.35 \times log_2 0.35\}]$$
$$= -[\{0.65 \times (-0.621)\} + \{0.35 \times (-1.515)\}] = -[\{-0.40365\} + \{-0.53025\}]$$
$$= -[-0.9339] = 0.9339 \approx 0.934$$

QUESTION 5: [2 marks]

What is the information gain if population density is selected as the root node of the model?

a. 0.119
b. 0.402
c. 0.134
d. 0.082

Correct Answer: d. 0.082

Detailed explanation: *Refer to lecture 33, slide 11.*

Total number of observations with population density as high = 7
For population density as high,
Total Number of observations with Yes for target variable (Bus stop needed) = 6
Total Number of observations with No for target variable (Bus stop needed) = 1

$$Entropy_{population\ density,high} = -\left[\left\{\frac{6}{7} log_2 \frac{6}{7}\right\} + \left\{\frac{1}{7} log_2 \frac{1}{7}\right\}\right]$$
$$= -[\{0.857 \times log_2 0.857\} + \{0.143 \times log_2 0.143\}]$$
$$= -[\{0.857 \times (-0.223)\} + \{0.143 \times (-2.806)\}] = -[\{-0.19111\} + \{-0.40126\}]$$
$$= -[-0.59237] = 0.59237 \approx \mathbf{0.592}$$

Total number of observations with population density as medium = 7
For population density as medium,
Total Number of observations with Yes for target variable (Bus stop needed) = 4
Total Number of observations with No for target variable (Bus stop needed) = 3

$$Entropy_{population\ density,medium} = -\left[\left\{\frac{4}{7} log_2 \frac{4}{7}\right\} + \left\{\frac{3}{7} log_2 \frac{3}{7}\right\}\right]$$
$$= -[\{0.571 \times log_2 0.571\} + \{0.429 \times log_2 0.429\}]$$
$$= -[\{0.571 \times (-0.808)\} + \{0.429 \times (-1.221)\}] = -[\{-0.46137\} + \{-0.52381\}]$$
$$= -[-0.98518] = 0.98518 \approx \mathbf{0.985}$$

Total number of observations with population density as low = 6
For population density as low,
Total Number of observations with Yes for target variable (Bus stop needed) = 3
Total Number of observations with No for target variable (Bus stop needed) = 3

$$Entropy_{population\ density,low} = -\left[\left\{\frac{3}{6} log_2 \frac{3}{6}\right\} + \left\{\frac{3}{6} log_2 \frac{3}{6}\right\}\right] = -[\{0.5 \times log_2 0.5\} + \{0.5 \times log_2 0.5\}]$$
$$= -[\{0.5 \times (-1)\} + \{0.5 \times (-1)\}] = -[\{-0.5\} + \{-0.5\}] = -[-1] = \mathbf{1}$$

$$Weighted\ Entropy_{population\ density} = \left[\left\{\frac{7}{20}\right\} \times \{0.592\}\right] + \left[\left\{\frac{7}{20}\right\} \times \{0.985\}\right] + \left[\left\{\frac{6}{20}\right\} \times \{1\}\right]$$
$$= [\{0.35\} \times \{0.592\}] + [\{0.35\} \times \{0.985\}] + [\{0.3\} \times \{1\}]$$
$$= [0.2072] + [\{0.3448\}] + [0.3] = \mathbf{0.852}$$

$$Information\ Gain_{population\ density} = [Entropy_{total}] - [Entropy_{population\ density}] = 0.934 - 0.852 = \mathbf{0.082}$$

QUESTION 6: [2 marks]

Consider the following statements with regards to the selection of attribute for root node among traffic volume, commercial activity and existing public transport, based on the information given above, and choose the correct option.

Statement A – For traffic volume as root node, the weighted entropy of the model is 0.402 and the information gain is 0.532.
Statement B – For commercial activity as root node, the weighted entropy of the model is 0.861 and the information gain is 0.073.
Statement C - For existing public transport as root node, the weighted entropy of the model is 0.773 and the information gain is 0.161.

a. The attribute commercial activity is most likely to be selected as root node as it has the highest weighted entropy among the three attributes.
b. The attribute traffic volume is most likely to be selected as root node as it has the highest information gain among the three attributes.
c. The attribute commercial activity is most likely to be selected as root node as it has the lowest information gain among the three attributes.
d. The attribute traffic volume is least likely to be selected as root node as it has the lowest weighted entropy among the three attributes.

Correct Answer: b. The attribute traffic volume is most likely to be selected as root node as it has the highest information gain among the three attributes.

Detailed Solution: *Refer to lecture 33, slide 11.*

QUESTION 7: [1 mark]

Which of the following statements correctly explains a limitation of decision trees and a common method used to address it?

a. Decision trees require feature scaling, which is addressed using normalization techniques.
b. Decision trees are prone to overfitting and instability, which can be reduced using pruning and ensemble methods like Random Forests.
c. Decision trees cannot handle categorical data, which is addressed by converting all features to numerical values.
d. Decision trees only work for regression problems, which is addressed using classification techniques.

Correct Answer: b. Decision trees are prone to overfitting and instability, which can be reduced using pruning and ensemble methods like Random Forests.

Detailed Solution: *Refer to lecture 33, slide 4.*

QUESTION 8: [1 mark]

In XGBoost, which hyperparameter specifically controls the fraction of training data randomly sampled to train each boosting tree, helping to reduce overfitting and improve generalization?

a. **learning_rate –** controls the step size of learning from each tree
b. **max_depth** – determines the depth of each tree
c. **subsample –** specifies the fraction of training instances used for each tree
d. **n_estimators –** defines the number of boosting rounds

Correct Answer: c. **subsample –** specifies the fraction of training instances used for each tree

Detailed Solution: *Refer to lecture 35, slide 7.*

QUESTION 9: [1 mark]
Which of the following correctly distinguishes Bagging, Random Forest, and Boosting (e.g., XGBoost)?

a. Bagging trains models sequentially to correct previous errors, Random Forest uses a single tree, and Boosting trains models in parallel.
b. Bagging trains multiple models on bootstrap samples in parallel, Random Forest additionally selects random subsets of features at each split, and Boosting trains models sequentially to reduce previous errors.
c. Bagging and Boosting both use feature randomness at each split, while Random Forest uses the full dataset without sampling.
d. Random Forest and Boosting both rely only on averaging predictions, while Bagging uses majority voting.

Correct Answer: b. Bagging trains multiple models on bootstrap samples in parallel, Random Forest additionally selects random subsets of features at each split, and Boosting trains models sequentially to reduce previous errors.

Detailed Solution: *Refer to lecture 35, slide 3,7 & 11.*

QUESTION 10: [2 marks]

A dataset contains 1,000 samples equally distributed across two classes (500 each). Two attributes are evaluated for splitting:

- Attribute A1 has 10 distinct values, each value uniquely identifies 100 samples belonging to only one class (perfectly pure subsets).
- Attribute A2 has 2 values, each splitting the dataset into 500 samples, with class distribution (400,100) and (100,400) respectively.

Which of the following statements is **CORRECT** regarding the selection of splitting attribute by Information Gain (IG) and Gain Ratio (GR)?

a. Both Information Gain and Gain Ratio will select A1 because it produces zero entropy after split.
b. Information Gain will select A1, but Gain Ratio may prefer A2 due to normalization using Split Information.
c. Information Gain will select A2, but Gain Ratio will select A1 due to lower split entropy.
d. Both Information Gain and Gain Ratio will select A2 because it produces more balanced partitions.

Correct Answer: a. Both Information Gain and Gain Ratio will select A1 because it produces zero entropy after split.

Detailed Solution: *Refer to lecture 33, slide 9 & 10.*
Initial entropy is maximum (since 500/500 split).

For Information Gain,
- A1 creates 10 pure subsets, so entropy after split = 0
- IG(A1) = maximum possible
- A2 creates two impure subsets, so entropy after split > 0
- IG(A2) < IG(A1)

So, Information Gain selects A1 attribute.

For Gain Ratio,
- Gain Ratio for attribute A1 = 0.301 *(Refer to Lecture 33, Slide 10)*
- Gain Ratio for attribute A2 = 0.278 *(Refer to Lecture 33, Slide 10)*

So, Gain Ratio also selects A1 attribute

<u>QUESTION 11:</u> [2 marks]

A parent node in a binary classification problem contains 100 samples with class distribution:
Class 1: 60
Class 0: 40
Two possible binary splits are evaluated using CART (Gini impurity).

<u>Split A</u>
- Left child: 30 (Class 1), 10 (Class 0)
- Right child: 30 (Class 1), 30 (Class 0)

<u>Split B</u>
- Left child: 50 (Class 1), 20 (Class 0)
- Right child: 10 (Class 1), 20 (Class 0)

Which of the following statements is **CORRECT**?

a. Split A is selected because it produces lower Gini in the left node.
b. Split B is selected because it gives larger weighted Gini reduction.
c. Split A is selected because it gives larger weighted Gini reduction.
d. Both splits produce the same weighted Gini, so CART chooses either randomly.

Correct Answer: b. Split B is selected because it gives larger weighted Gini reduction.

Detailed Solution: *Refer to lecture 34, slide 4.*

Parent Gini
Gini_parent = 1 - (0.6^2 + 0.4^2) = 1 - (0.36 + 0.16) = 1 - 0.52 = 0.48

For Split A,

Left Node (40 samples: 30,10)
Gini_Left_Node_1 = 1 - (0.75)^2 + (0.25)^2 = 1 - (0.5625 + 0.0625) = 1 - 0.625 = 0.375

Right Node (60 samples: 30,30)
Gini_Right_Node_1 = 1 - (0.5^2 + 0.5^2) = 1 - (0.25 + 0.25) = 0.5

Weighted Gini (Split A) = Gini_A = (0.40) X (0.375) + (0.60) X (0.50) = 0.15 + 0.30 = 0.45
Reduction (Split A) = 0.48 - 0.45 = **0.03**

For Split B,

Left Node (70 samples: 50,20)
Gini_Left_Node_2 = 1 - (0.71)^2 + (0.29)^2 = 1 - (0.5041 + 0.0841) = 1 - 0.5882 = 0.4118

Right Node (30 samples: 10,20)
Gini_Right_Node_2 = 1 - (0.33)^2 + (0.67)^2 = 1 - (0.1089 + 0.4489) = 1 - 0.5578 = 0.4422

Weighted Gini (Split B) = Gini_B = (0.70) X (0.4118) + (0.30) X (0.4422) = 0.2883 + 0.1327 = 0.4210
Reduction (Split B) = 0.48 - 0.4210 = **0.0590**

Since Split B produces larger impurity reduction, CART selects it.

QUESTION 12: [2 marks]
Consider the following codes given below and choose the correct option.

Code A
```
from sklearn.model_selection import GridSearchCV
param_grid = {"fit_intercept":[True, False]}
grid = GridSearchCV(model, param_grid, cv=5)
grid.fit(X, y)
```

Code B
```
from sklearn.model_selection import cross_val_score
scores = cross_val_score(model, X, y, cv=5)
print(scores.mean())
```

a. Code A performs systematic hyperparameter tuning and Code B performs cross-validation that gives reliable estimate for small datasets.
b. Code A performs cross-validation that gives reliable estimate for small datasets and Code B performs systematic hyperparameter tuning.
c. Code A handles missing values and scaling together and Code B builds model without data leakage.
d. Code A builds model without data leakage and Code B handles missing values and scaling together.

Correct Answer: a. Code A performs systematic hyperparameter tuning and Code B performs cross-validation gives reliable estimate for small datasets.

Detailed Solution: *Refer to lecture 32*

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